

Data Aggregation in R: Summarizing Data for Reports and Charts

Why Aggregation Is Different

Every other transformation in the `transf` collection — sampling, cleaning, scaling, encoding, smoothing, balancing, feature selection, dimensionality reduction — preserves the row-level structure of the dataset: each output row still corresponds to one input row.

Collapses Rows

Aggregation collapses rows. The output has one row per group, not per original observation.

Feeds Reporting

Useful before plotting and reporting. Exactly the kind of "one number per category" summary that feeds a bar chart or a stakeholder table.

Closes the Pipeline

Closes the preprocessing pipeline. Sampling, cleaning, scaling, and encoding prepare individual rows for modeling; aggregation prepares grouped rows for summarizing.

Grouped Summaries – aggregation

Summarizes a dataset by groups through named aggregation expressions. The interface follows the same "configure, then transform" pattern as every other transf technique, borrowing directly from dplyr-style summary syntax.

```
agg <- aggregation(  
  "Species",  
  mean_sepal = mean(Sepal.Length),  
  sd_sepal   = sd(Sepal.Length),  
  n          = n()  
)
```

```
iris_agg <- transform(agg, datasets::iris)
```

- **First argument:** the grouping attribute ("Species").
- **Remaining arguments:** named expressions evaluated within each group — any summary function works (mean, sd, n(), median, custom).
- **Output:** one row per group, one column per named expression.

What to Observe

- **Dataset-level restructuring:** aggregation is not a row-level edit — a conceptual category of its own inside the `transf` collection.
- **Scales cleanly:** the named-expression interface scales to any number of summary statistics without changing the call shape.
- **Natural next step:** feed the summarized table directly into the graphics comparison charts (bar, grouped bar, lollipop).



Acknowledgements



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